

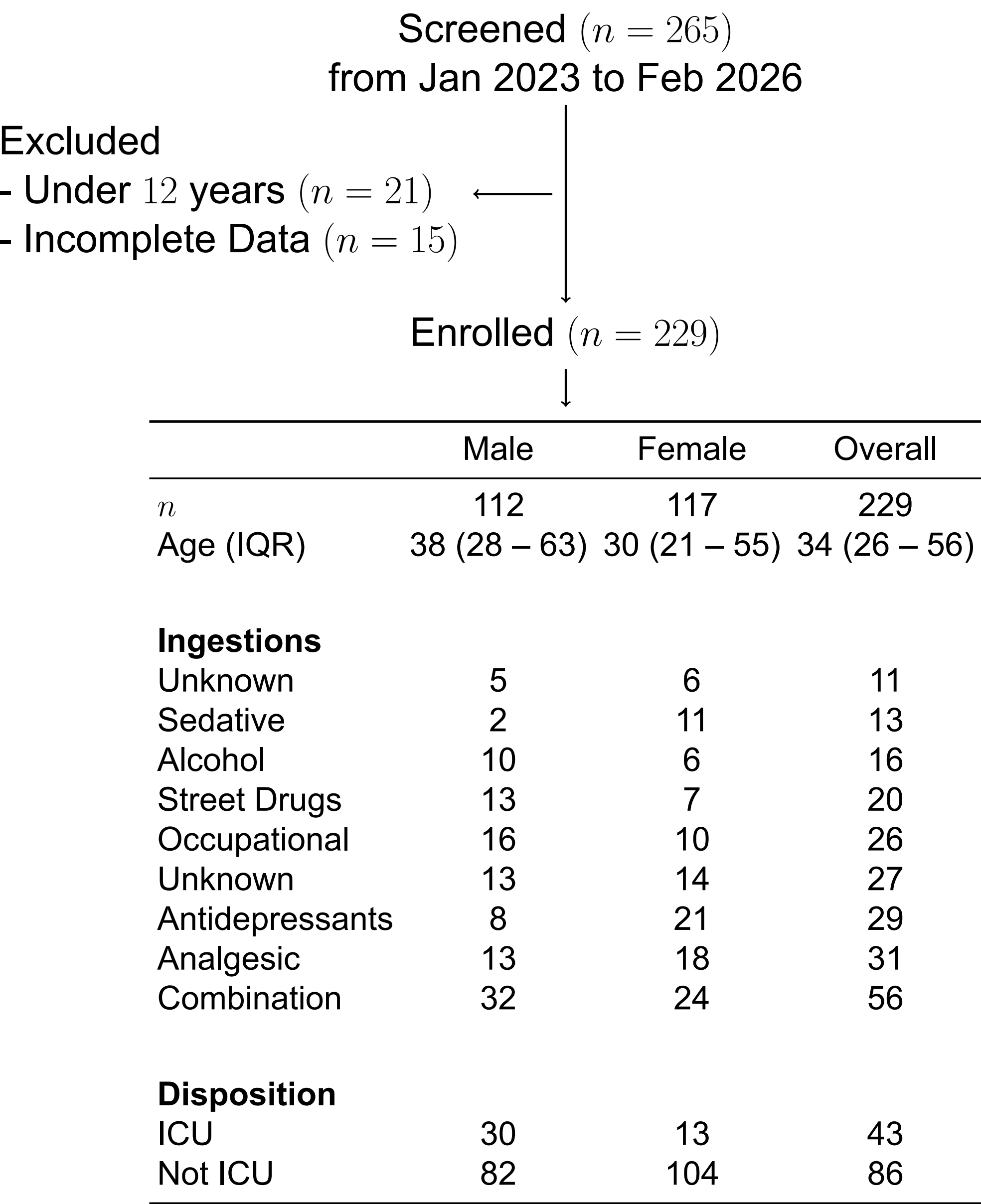
Biological Sex Helps Predict Disposition of Symptomatic Poisoned Patients in the ED but not in the ICU.

Predictive Value of Biological Sex for ICU-Level Care

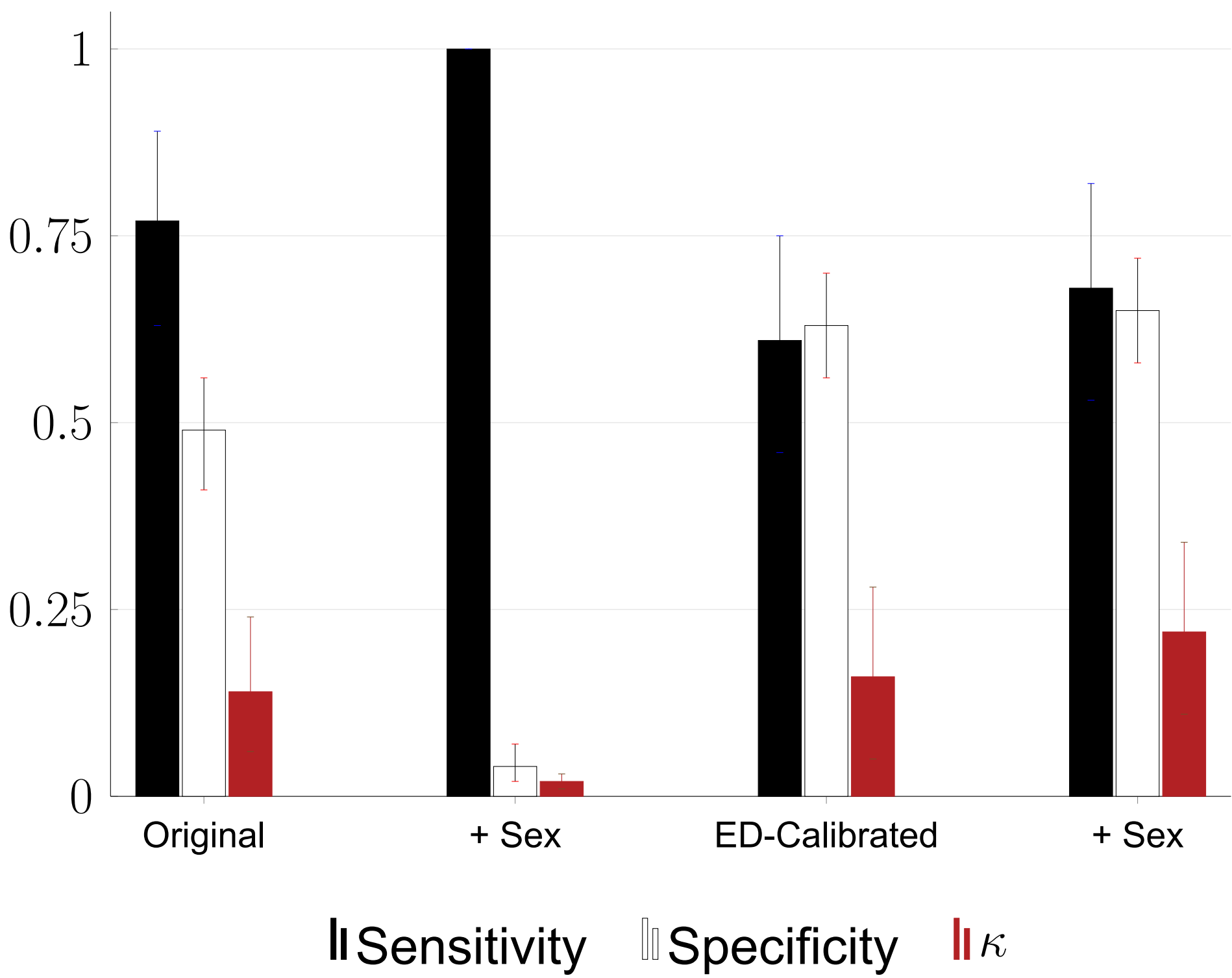
Background

- Nearly 25% of poisoned patients admitted to the hospital do not require treatment.
- The ICU Requirement Score may decrease unnecessary ICU admissions from the ED.
- Its ED disposition predictions do not always agree with bedside toxicologists.
- The IRS does not include biological sex, despite its association with poisoning outcomes.

Our goal was to improve agreement of IRS with toxicologist-recommended disposition by adding biological sex.



Model Performance



Agreement with Toxicologists

ED-Calibrated Model				+ Biological Sex			
	ICU	Not ICU	Total		ICU	Not ICU	Total
ICU	26	69	95	ICU	29	65	94
Not ICU	17	117	134	Not ICU	14	121	135
Total	43	186	229	Total	43	186	229

Methods

- Secondary analysis of one institutional toxicology registry
- Refit logistic regression refit with and without biological sex.
- Outcome: Inter-rater between model and toxicologist recommendation

Conclusions: Including biological sex improves predicting disposition of symptomatic poisoned patients on initial presentation in the ED, but not once in the unit. This may reflect biological sex functioning as a proxy for dose and ingestion subtype.

Next Steps

- Including dose | Develop ICU vs Floor vs Home/Psych Model ⇒ Prospective Test